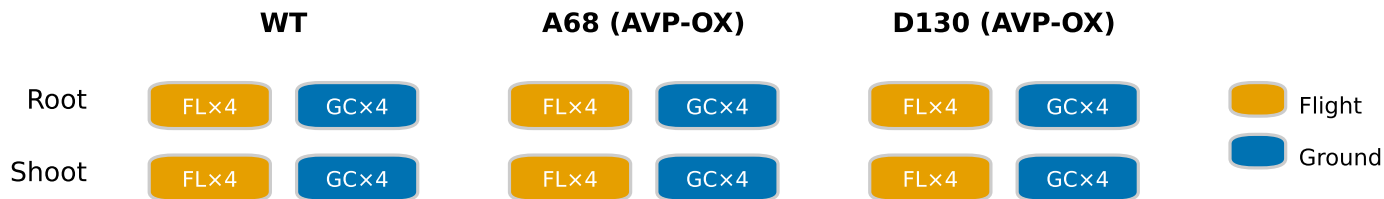


Fig 1. TIC-TOC experimental design and analysis workflow

AVP-OX cotton (*Gossypium hirsutum*) aboard the ISS (SpaceX CRS-22) vs matched ground controls

A. Full factorial design — 4 replicates/group, 48 RNA-seq libraries



B. Two paired readouts

Root architecture

days 3-6 imaging → SmartRoot → RSML · 198 tracings, 53 plants

Transcriptome

root + shoot RNA-seq · UTX-TM1 v2.1 · 59,918 genes · DESeq2

C. Analysis workflow



D. Root imaging time series

